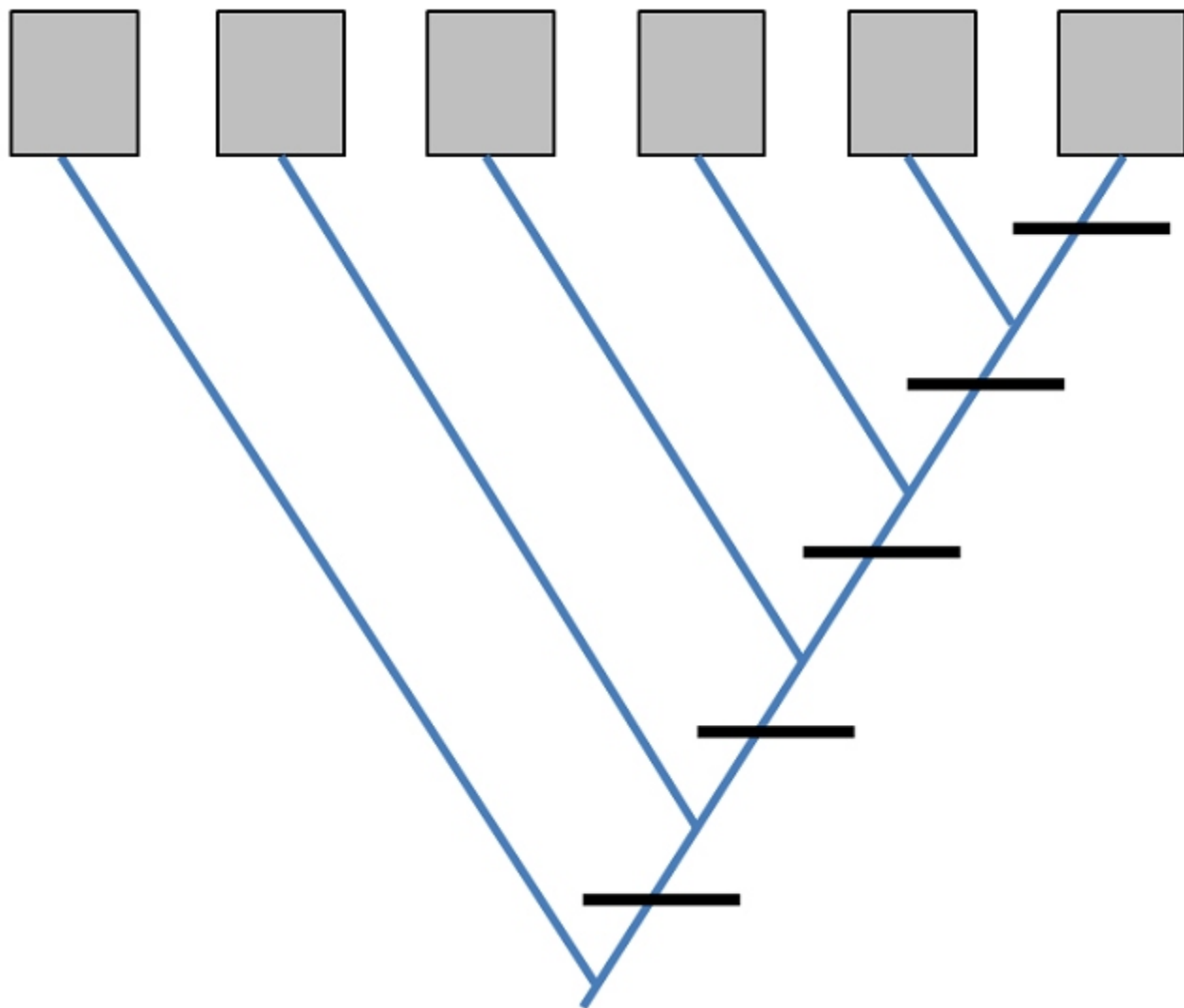




|             | Thick Hooves<br>(1) | Big Teeth<br>(2) | Soft Fur<br>(3) | Squeaky Voice<br>(4) | Red Nose<br>(5) |
|-------------|---------------------|------------------|-----------------|----------------------|-----------------|
| Dasher (D)  | Y                   | N                | Y               | N                    | N               |
| Prancer (P) | N                   | Y                | N               | Y                    | N               |
| Vixen (V)   | N                   | Y                | N               | N                    | N               |
| Comet (C)   | Y                   | N                | N               | N                    | N               |
| Rudolph (R) | Y                   | N                | Y               | N                    | Y               |
| Blitzen (B) | Y                   | Y                | N               | N                    | N               |

Based on the information in the table below, construct a cladogram (or family tree) that depicts the relatedness of Christmas reindeer based on ancestral or derived characters. **Place each of the reindeer "D, P, V, C and B" in the gray boxes on the terminal branches of the tree.** Also, **place the character state using the character numbers (1-5) on the black dashes which intersect the tree.** Be sure to indicate whether these characters **APPEARED** or **DISAPPEARED** at that intersection on the tree using a + or – symbol (i.e. 1+ or 2-).

\*\*\* For cladistic purposes, Rudolph will be considered the OUTGROUP since he is the OLDEST

